



Expert feedback report

Phase 1 Round 1

Dear expert,

Thank you for participating in Phase 1, Round 1 of the ECTO project, and for the thoughtful suggestions and comments you provided.

This completed report summarizes how the expert panel evaluated the Phase 1 list of theorists' core tasks and responsibilities after inclusion of all 13 completed responses. The quantitative survey results are summarized below, followed by item-level comments and responses to the main revision suggestions.

Consensus rule and updating

During each round, an item is designated as **consensus reached** when its mean rating is ≥ 4.5 and no more than 25% of valid respondents rate it 3 or lower. The item is then dropped from the next consensus round. Thus, once consensus has been reached on an item, it will not be reevaluated in subsequent rounds.

Written recommendations are used to revise, merge, split, add, or remove items. These decisions are made by the project lead. If you disagree with a decision, please let us know in response to this report.

Round summary

In this round:

- 13 experts completed the survey.
- 8 items were evaluated.
- 8 items reached consensus and will be dropped from the next consensus round.
- 0 items failed to reach consensus under the predefined rule.
- 27 item-level feedback responses were provided.
- 9 responses suggested additional items.
- 11 responses discussed important but delegable tasks.
- 9 general comments were provided.



Because all eight items reached consensus, the eight-item structure has been retained. The written comments were used to make minimal wording changes, reduce ambiguity, and clarify the distinction between core theoretical responsibility and delegable execution.

Some new items were suggested, but others said that the list of core responsibilities was already quite long. Thus, suggested additional items were incorporated into existing items where possible. Suggestions that concerned technical, empirical, infrastructural, applied, or communicative execution were placed in the section **Important but delegable tasks** below. Your comments were also used to update the introduction of the list of core responsibilities of theorists. This clarification adds that the eight items describe core theoretical responsibilities, not tasks that every theorist must personally execute alone. Many tasks require collaboration with empirical researchers, measurement experts, statisticians, computational modelers, domain specialists, designers, or research software engineers. The theorist's non-delegable responsibility is to preserve the theoretical meaning of the work: what is being explained, represented, measured, tested, compared, revised, or applied.

If you have any questions, please do not hesitate to reach out.

Sincerely,
Noah van Dongen

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Item-level decisions

Item 1: Identification and formation of explananda

Original item:

The theorist identifies, categorizes, and orders the phenomena or capacities to be explained by theory. The theorist distinguishes robust phenomena that warrant explanation from raw data patterns and fleeting effects (e.g., potential false positives) that require further empirical research.

Survey result:

- Valid respondents: 13
- Mean rating: 5.62
- Percentage rating 3 or lower: 0.0%
- Consensus status: Consensus reached
- Recommendation categories: 1 revise wording/description; 1 split into multiple items

Comments:

- I would recommend revising the item, because early theory development may also legitimately address emerging, puzzling, or weakly understood patterns, as long as their epistemic status is made explicit. A phenomenon does not need to be fully robust before it can guide theorizing, but the theory should distinguish between well-established explananda, provisional empirical patterns, and speculative targets. Theories do not have to be the endpoint of a research process, but could be an invitation/startingpoint for revision and further development.
- I fully agree with the first sentence. It concerns the terminology and conceptual framework of a theory, or more specifically, how a phenomenon is defined and used within a theory. The second sentence, however, concerns a different issue: the distinction between the terminology or conceptual framework of a theory and data, understood as something external to the theory of interest. This distinction is important, but these are two separate matters. In my view, the second sentence is more difficult to address because it includes terms such as "robust" and "raw" data patterns. How can one know whether a phenomenon is robust? To judge the robustness of a phenomenon, researchers must have observed relevant patterns in the data repeatedly over time; that is, there must be a collection of facts that is consistent with what the theory claims. Moreover, researchers must agree that this collection of facts counts as evidence for the robustness (instability) of the phenomenon. For this reason, the second sentence is difficult to apply in practice. It is not clear how researchers can reliably distinguish between a "robust phenomenon" and "raw data patterns."



Additional comments addressed in this item:

- **Suggested new items, Comment 2:** There is a tradition of theorizing as a problem solving exercise (advanced by Larry Laudan). The theorist should specify what problem the theory is solving.

Revised item:

The theorist identifies, categorizes, and orders the phenomena, capacities, or problems to be explained by theory. The theorist distinguishes established explananda from provisional patterns, speculative targets, and raw data patterns, while making their epistemic status explicit.

Explanation:

Consensus reached. The item was minimally revised to allow established as well as provisional or emerging explananda, provided their epistemic status is explicit, and to include the idea that theorizing can be oriented toward solving a problem.

Item 2: Developing and evaluating construct architecture

Original item:

The theorist defines, differentiates, coordinates, and, where necessary, prunes constructs; builds ontologies and taxonomies; and specifies how theoretical terms refer to psychological phenomena, capacities, attributes, or processes.

Survey result:

- Valid respondents: 13
- Mean rating: 6.00
- Percentage rating 3 or lower: 0.0%
- Consensus status: Consensus reached
- Recommendation categories: none

Comments:

No item-level comments were provided.

Additional comments addressed in this item:

No end-of-survey comments were primarily addressed in this item.

Revised item:

The theorist defines, differentiates, coordinates, and, where necessary, prunes constructs; builds



ontologies and taxonomies; and specifies how theoretical terms refer to psychological phenomena, capacities, attributes, states, processes, or mechanisms.

Explanation:

Consensus reached. The item was retained with only minor wording alignment by adding "states" and "mechanisms" as possible theoretical referents.

Item 3: Coordinating the relation between theory and measurement

Original item:

The theorist specifies how constructs could, in principle, be detected, measured, or behaviorally evidenced; what counts as valid indicators; and what assumptions connect measures to theoretical entities.

Survey result:

- Valid respondents: 13
- Mean rating: 5.54
- Percentage rating 3 or lower: 7.7%
- Consensus status: Consensus reached
- Recommendation categories: 1 revise wording/description

Comments:

- Some constructs are not "detected" or "measured" in the ordinary meaning of these words; for instance, constructs that act as inductive summaries (how many Roman Emperors do you know?) formative measures (SES) or indices (Dow Jones). These constructs are legitimate but do not act as "detectable" attributes because the causal direction does not obey attribute->measure.

Additional comments addressed in this item:

- **Important but delegable tasks, Comment 1, part #2:** #2 the theorist should define the measurement problem - measurement experts could solve parts of the technical implementation (i.e., scale construction, item response modeling, reliability estimation, measurement invariance testing, and validation)
- **Important but delegable tasks, Comment 6:** I rated 3. Coordinating the relation between theory and measurement lower than the other tasks. I think it requires strong empirical experience to derive the most valid measurement techniques. People specializing in theoretical psychology may not necessarily have such experience. When point 2. before is done well, I suspect that



measurement is already constrained to such an extent, that specific measurement proposal don't have to be made. Moreover, what if our empirical techniques improve over time. Then, this former proposal is outdated. Hence, I agree that point 3. is important, yet it could be delegated to experimentalists.

- **Important but delegable tasks, Comment 7:** Yes building tests that manifest good construct validity

Meaning that the test empirically behaves in ways that align with key theoretical propositions postulated by the theory

- **General comments, Comment 5:** Be sure to include that theories can be developed for and evaluated in diverse settings including naturalistic experimental and clinical

And using a variety of study designs

To evaluate theories of constructs typically construct valid measures are required

Evaluation of cons

Revised item:

The theorist specifies how constructs could, in principle, be operationalized, indexed, measured, or otherwise evidenced; what counts as valid indicators or indices; and what assumptions connect these to theoretical entities. Technical measurement development can be collaborative or delegated.

Explanation:

Consensus reached. The item was minimally revised to avoid implying that all constructs are detectable attributes; it now covers reflective indicators, formative measures, indices, and other operational representations, while retaining the distinction between theoretical responsibility and delegable technical measurement development.

Item 4: Proposing, specifying, and connecting mechanisms

Original item:

The theorist proposes and refines mechanisms, distinguishing sketches from more complete schemata and relating psychological explanations to neural, bodily, social, developmental, or environmental levels without reducing one level to another.

Survey result:

- Valid respondents: 13
- Mean rating: 5.15
- Percentage rating 3 or lower: 15.4%
- Consensus status: Consensus reached
- Recommendation categories: 4 revise wording/description



Comments:

- "Mechanism" is not a single concept by far and even if it's accepted that it means all the things it could, I think it still may be an optional part. Somebody may want to be functionalist in a way that excludes current mechanistic analysis. Perhaps we can talk more at some point over voice (because it's a can of worms) or I can send you some work I have under development on this -- I think it may help. However, I am also open to interpreting the above as if not prefaced with a "should" or "must".
- In all definitions, so far, I find there to be quite a lot of jargon. That, at least to me, makes the item difficult to read (had to read a couple of times). One reason may be that the sentences are quite long (also in this case). For this particular item: what is considered a mechanism/ when is something a mechanism? I feel this word is been used *a lot* and the more I think about it the less certain I am what we refer to/ whether there is consensus in to what we consider as mechanisms.
- While I think reductionism rarely works, it does work sometimes! The reduction of temperature to average kinetic energy is a pretty cool feat. So I feel it's weird to rule out reduction a priori.
- First, there are several words/concepts that require an explanation: what is meant by "sketch"? What is a "schema" and when is a schema "complete"? Second, I was not fully sure whether I understood the term "level" here: I think I understand the difference between intraindividual, interindividual, interpersonal, intergroup level (this is the taxonomy of levels I know from social psychology). But I don't really see how "developmental" or "environmental" fits into that picture. Also, I was missing the intraindividual-cognitive/motivational/emotional level in the list. Third, I wondered whether "reducing one level to another" is necessarily a bad thing -- sometimes, such "inter-level redundancies" can be informative. For instance, an "objective" injustice (on the social/interpersonal level) manifests in anger/outrage (on the intrapersonal level), which results in aggressive behavior (intrapersonal), which affects others (interpersonal). The term "reduce" may be misunderstood in terms of "choose one level for each possible mechanism", which may not always work...

Additional comments addressed in this item:

- **Important but delegable tasks, Comment 8, part #1:** * Many of the things in the list can (in my view) only be achieved in collaboration with other researchers -- for instance, judging what counts as a valid indicator for a construct (you may need a test developer, diagnostician, methodologist here); relating potential mechanisms to each other (you may need a psychologist for the intraindividual level, a sociologist for the macrosocial level, an anthropologist for the evolutionary level, etc.). So many of the things here cannot be done by the theorist alone.

Revised item:

The theorist proposes and refines mechanistic, functional, or process-based explanations, distinguishing preliminary sketches from more specified accounts and relating psychological explanations across



cognitive, affective, motivational, behavioral, bodily, neural, social, developmental, and environmental levels.

Explanation:

Consensus reached. The item was revised to reduce jargon, broaden mechanism language to include functional and process-based explanation, clarify the relevant levels of explanation, and avoid ruling out reduction or informative inter-level relations a priori.

Item 5: Constructing theories and their formal representations (models)

Original item:

The theorist builds verbal theories and formal models (e.g., diagrammatic, computational, mathematical, etc.) that encode explanatory principles, clarify assumptions, and enable systematic comparison.

Survey result:

- Valid respondents: 13
- Mean rating: 5.23
- Percentage rating 3 or lower: 15.4%
- Consensus status: Consensus reached
- Recommendation categories: 3 revise wording/description; 1 other

Comments:

- Nuance might be needed on what is happening here. So yes, absolutely there are informal theories but there are formal theories that are not models. Can say more if needed, but perhaps this is enough: Guest, O. & Martin, A. E. (2021). How Computational Modeling Can Force Theory Building in Psychological Science. *Perspectives on Psychological Science*.
<https://doi.org/10.1177/1745691620970585>
- It is not clear what exactly is meant with explanatory principles? I see that this is an important feature, but I worry that researchers will interpret this aspect in different ways.
- Model enables and facilitates empirical testing. Model illustrates and highlights core theoretical propositions including theorized dimensions and relations to outside phenomena including correlates predictors causes consequences moderators and mediators
- I am not 100% sure about verbal theories and formal models, maybe or instead of and?



Additional comments addressed in this item:

- **Suggested new items, Comment 1, part #2:** #2 The theorist creates theories that are usable by other researchers, including clear verbal specifications, proposition lists, diagrams, model variants, construct definitions, and documentation of assumptions, uncertainties, and open questions. --> its kind of in your points - but I would love it to be very explicit.
- **Suggested new items, Comment 3:** I feel there was none of b and d from Table 1 in Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good?. Computational Brain & Behavior. <https://doi.org/10.1007/s42113-023-00193-2>
- **Suggested new items, Comment 5:** Communicating the theory properly, also to non-theorists. Theories should eventually inform other psychologists' work, yet many have limited experience with theoretical work, especially formal modeling. It would be problematic if theoretical psychology develops independent from mainstream psychology. Just imagine a typical psychologists having to engage with a mathematical model relying on a set of differential equations. To me, there is an inherent challenge in presenting the necessary details of theoretical work and not overwhelming the readers one wants to reach. Navigating this challenge, in my opinion should be part of a theorist's tasks.
- **Important but delegable tasks, Comment 1, part #1:** #1 the theorist must know what is being formalized, but does not necessarily have to code or mathematically formalize the model. The theorist task is to ensure that the model faithfully represents the theory.
- **Important but delegable tasks, Comment 3:** Both formal proofs and programming spring to mind, but this might be too limited in how we see "theorist" if those to which they are delegated are not theorists.
- **Important but delegable tasks, Comment 10:** Formalizing

Revised item:

The theorist builds verbal, formal, diagrammatic, computational, mathematical, or hybrid theories, models, and model families that specify core propositions, clarify assumptions, support comparison and testing, and remain usable by relevant scientific audiences.

Explanation:

Consensus reached. The item was revised to avoid conflating theories, models, and formal representations, and to emphasize that theories should be usable by relevant scientific audiences.

Item 6: Exploring and repairing existing theories

Original item:

The theorist extends theories, deepens them, resolves internal tensions, adds mechanisms, examines



analogies, develops variants, and constructs how-possibly models when this improves coherence, richness, scope, or future testability.

Survey result:

- Valid respondents: 13
- Mean rating: 4.85
- Percentage rating 3 or lower: 7.7%
- Consensus status: Consensus reached
- Recommendation categories: 4 revise wording/description; 1 other

Comments:

- Maybe we could add "when this improves ... precision & explanatory value"
- You should add "constrains" not just "extends" -- see work like: Guest, O., Blokpoel, M., & van Rooij, I. (2026). What the Func? Multiple Realizability Need not be Vague. Zenodo. <https://doi.org/10.5281/zenodo.19388964>
- I would like to add that the specification and construction of new theories should involve situating any new theory within the context of existing theories, or using it to resolve anomalies, inconsistencies, or incoherence within an existing theory network, which has been partly included in above paragraph. It is important to note that there is no single theory that explains everything; instead, theories build on other formal or informal theories. One problem in psychology is that its theoretical landscape and terminology remains fragmented, lacking broad and integrative theories found in fields such as physics or mathematics. This point is important because the primary goal should not to create further isolated "theory islands" in psychology, but rather to develop "bigger" theories that are meaningfully connected to broader theoretical networks.
- Can sentence be split into two? Can "how-possibly models" be rephrased?
- Need an explanation somewhere of the core tasks or functions of theory including description explanation prediction/hypothesis generation, control/guide interventions, building useful models fruitfulness (generates novel insights and predictions) and esthetics

Additional comments addressed in this item:

- **Suggested new items, Comment 6:** See earlier points regarding the functions of theory Theories can be evaluated according to the same criteria for which they were originally developed eg description explanation prediction control fruitfulness
- **General comments, Comment 1:** Some items are extremely dense. Good theory development often requires narrowing, rejecting or specifying expected failures. I would therefore consistently include criteria such as precision and parsimony



Revised item:

The theorist extends, constrains, deepens, and repairs theories by resolving tensions, refining mechanisms, examining analogies, developing variants, and situating theories in relation to other theories when this improves coherence, precision, parsimony, explanatory value, scope, or testability.

Explanation:

Consensus reached. The item was revised to add constraining as well as extending theories, to remove "how-possibly models," and to include precision, parsimony, explanatory value, and relation to other theories as criteria for disciplined exploratory work.

Item 7: Derivation, explanatory appraisal, scope mapping, and comparison

Original item:

The theorist derives implications from theories and models, including explanations of established phenomena, observational predictions, intervention predictions, counterfactual expectations, and expected failures. The theorist identifies auxiliary assumptions, specifies boundary and scope conditions, compares rival theories, conducts anomaly analysis, evaluates how well theories account for existing evidence, and makes explicit what evidence should count for or against which version of a theory.

Survey result:

- Valid respondents: 13
- Mean rating: 4.77
- Percentage rating 3 or lower: 23.1%
- Consensus status: Consensus reached
- Recommendation categories: 2 revise wording/description; 4 split into multiple items; 1 other

Comments:

- The content is good, but it may be too packed. I would also emphasize that theorists should distinguish between implications that follow from the core theory and implications that depend heavily on auxiliary assumptions.
- Add "qualitative predictions" -- see for why and what I mean: Guest, O. Martin, A. E., & van Rooij, I. (2026). On Models, Prediction, and Scientific Roles Thereof: Commentary on Sanbonmatsu et al. (2025). Zenodo. <https://doi.org/10.5281/zenodo.19983695>
- I'm not sure I agree with the first 'role', as I think that implications and explanations are quite domain-dependent and that there will be a lot of different ways such a thing could go. I think, therefore, that that role is not (only) the job of the theorist. The second part about identifying



assumptions and boundary conditions is a somewhat separate role, and while it can be related to the first I think it makes more sense to separate it. Not least, from my perspective, because I have mixed feelings about the first to begin with, but do agree that the second is a role of the theorist. So to have them together makes my response to them quite neutral on balance which masks the nuance that's there.

- Not sure this sounds like an activity in Theorizing, more in using theories
- A lot of things are covered here! Why are these grouped under one item? If there's a reason, can't the item have a more overarching title?
- I do think that everything included in this list here makes perfect sense, but my hunch is that this is a quite broad category, broader than the others, so it may make sense to split it into two or even three sub-categories: (1) identify auxiliary assumptions and boundary conditions (to make the theory / its predictions actually testable), (2) identify cases in which the theory / its predictions have a high chance to be falsified or qualified, (3) keep track of the state of evidence against or in favor of a theory / its predictions.
- Too vague and rambling Lead off with theory facilitates comparisons contrasts and evaluation of core propositions in ways that align with its functions. For example descriptive power is measurable via the size and extent of the glossary Explanatory power is measurable via the number of phenomena the theory can account for including regularities irregularities correlation prediction causation moderation and mediation Predictive power is measurable via classification accuracy variance explained and prediction of rare events/irregularities Control is measurable via development of effective efficient interventions including effect size sessions required to effect sufficient change number of reliable improvers vs number of treatment nonresponders and reliable deteriorators creative/ novel insights for treating refractory cases and developing coherent complex interventions that integrate many elements to treat complex cases eg multisystemic treatment refractory intergenerational cases

Additional comments addressed in this item:

- **Suggested new items, Comment 4:** I think there's an argument for the theorist also having a distinct role in evaluating the work of other theorists. I don't think theorists work in silos (or think that they should not), and to be working with others, collaborating but also evaluating the work of peers should be an explicit role as it is in other domains of research.
- **Suggested new items, Comment 6:** See earlier points regarding the functions of theory Theories can be evaluated according to the same criteria for which they were originally developed eg description explanation prediction control fruitfulness
- **General comments, Comment 1:** '- some items are extremely dense
- good theory development often requires narrowing, rejecting or specifying expected failures. I would therefore consistently include criteria such as precision and parsimony
- **General comments, Comment 4:** Overall, I think the lists is very nice and helpful as a guide. However, beyond the task I think is missing (see above), my impression was that points 7. and 8., as currently formulated, are at least partly redundant. To me "deriving implications" and "makes



explicit what evidence should count for or against which version of a theory" (both 7) are similar to "formulating a theory-guided research program" (8). Moreover, "evaluates how well theories account for existing evidence" (7) sounds very similar to "evaluates whether existing theories can account for known phenomena" (8). My suggestion would be to either merge these two points or demarcate them more.

- **General comments, Comment 6:** I do like that list very much. But my feeling after reading the list of task was: For some of these tasks, a "division of labor" may be helpful or even necessary. See my comment above: some tasks require expert knowledge that the "theorist" may not have. In addition, I was wondering whether a "theorist" may sometimes take the role of a "critical friend", who, in adversarial collaboration with a subject-matter researcher, works towards refining a theory specification up to the point where the theory is actually testable. Doing this in an "adversarial collab" style may help prevent the tendency to immunize your theory-testing activity...

Revised item:

The theorist derives implications from theories and models, including explanations of established phenomena, qualitative or quantitative predictions, intervention predictions, counterfactual expectations, and expected failures. The theorist identifies auxiliary assumptions, maps scope and boundary conditions, compares rival theories, and specifies what evidence would count for or against each theory.

Explanation:

Consensus reached. The item was retained as one item but shortened and clarified by distinguishing explanation of established phenomena, qualitative or quantitative prediction, intervention prediction, counterfactual expectation, expected failure, auxiliary assumptions, scope, comparison, and evidence appraisal.

Item 8: Integration, revision, and programme design

Original item:

The theorist integrates fragmented empirical findings into coherent explanatory structures, evaluates whether existing theories can account for known phenomena, and revises, forks, merges, or discards theories in light of accumulated evidence. The theorist formulates theory-guided research programmes that organize future observational, experimental, interventional, formal, and measurement work. This includes articulating standards by which theories and models should be judged.



Survey result:

- Valid respondents: 13
- Mean rating: 5.31
- Percentage rating 3 or lower: 15.4%
- Consensus status: Consensus reached
- Recommendation categories: 2 revise wording/description; 1 split into multiple items; 1 other

Comments:

- Would love to make clear, that "accumulated evidence" should include not only empirical findings, but also evidence about measurement validity and failed derivations
- Sounds like meta*research, not Theorizing
- Two things: First, I am not sure whether it's the theorists job to "evaluate whether existing theories can account for known phenomena". Second, formulating theory-guided research programmes is probably a separate point.
- Theorist approaches theory development as a process of ongoing evaluation and evolution either refining capacity to perform desired functions or when needed engages in a paradigm shift per Kuhn

Additional comments addressed in this item:

- **Suggested new items, Comment 1, part #1:** #1 The theorist documents the origine, versioning, and revision history of theories, including which propositions, constructs, mechanisms, scope conditions, and auxiliary assumptions have been accepted, revised, forked, or discarded over time.
- **Suggested new items, Comment 3:** I feel there was none of b and d from Table 1 in Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good?. Computational Brain & Behavior. <https://doi.org/10.1007/s42113-023-00193-2>
- **Suggested new items, Comment 7:** I would include the following: "Keep track of all activities, decisions, procedures, including all empirical evidence that has been conducted with the aim of testing a theory. This should include any failed attempts. To 'keep track of' also means to document everything that has been done, preferably in a FAIR way." (FAIR = findable, accessible, interoperable, reusable). Reason: necessary to enhance transparency and openness regarding the theorist's work.
- **Important but delegable tasks, Comment 1, part #3:** #3 theorists design or guide theory-relevant research programmes, but they do not always have to run the empirical studies themselves
- **Important but delegable tasks, Comment 5:** Building infrastructure for archiving and reusing theory
- **Important but delegable tasks, Comment 8, part #2:** * "Formulating theory-guided research programmes" (which was one point in task no. #8) could be delegated to other experts.



- **General comments, Comment 2:** The theorist has the choice of retaining a theory after it has been "falsified." The principle of tenacity matters and even Popper agreed that theorists should not be hasty in giving up theories. Relatedly, Larry Laudan also proposed a logic of pursuit, which indicates when a research direction is worthy of allocation resources to.
- **General comments, Comment 4:** Overall, I think the lists is very nice and helpful as a guide. However, beyond the task I think is missing (see above), my impression was that points 7. and 8., as currently formulated, are at least partly redundant. To me "deriving implications" and "makes explicit what evidence should count for or against which version of a theory" (both 7) are similar to "formulating a theory-guided research program" (8). Moreover, "evaluates how well theories account for existing evidence" (7) sounds very similar to "evaluates whether existing theories can account for known phenomena" (8). My suggestion would be to either merge these two points or demarcate them more.
- **General comments, Comment 5:** Be sure to include that theories can be developed for and evaluated in diverse settings including naturalistic experimental and clinical
And using a variety of study designs
To evaluate theories of constructs typically construct valid measures are required
Evaluation of cons

Revised item:

The theorist integrates fragmented empirical, formal, and measurement-related evidence; revises, retains, forks, merges, or discards theories; documents assumptions, versions, and open questions; and helps design theory-guided research programmes. This includes articulating standards for judging theories, including their explanatory value, empirical interface, usability, and potential harms.

Explanation:

Consensus reached. The item was revised to clarify that accumulated evidence includes empirical, formal, and measurement-related evidence, and to add theory retention, documentation, programme design as partly collaborative work, and standards including usability and potential harms.



Suggested new items

Survey question:

Should any items be added to the list of theorists' core responsibilities/tasks? If so, please describe them and give at least one argument for their inclusion.

Comment 1: #1 The theorist documents the origine, versioning, and revision history of theories, including which propositions, constructs, mechanisms, scope conditions, and auxiliary assumptions have been accepted, revised, forked, or discarded over time. #2 The theorist creates theories that are usable by other researchers, including clear verbal specifications, proposition lists, diagrams, model variants, construct definitions, and documentation of assumptions, uncertainties, and open questions. --> its kind of in your points - but I would love it to be very explicit.

Response 1: Addressed in Items 5 and 8: usability is added to theory construction, while documentation of assumptions, versions, and open questions is added to integration and revision.

Comment 2: There is a tradition of theorizing as a problem solving exercise (advanced by Larry Laudan). The theorist should specify what problem the theory is solving.

Response 2: Addressed in Item 1 by adding "problems" as possible targets of theorizing.

Comment 3: I feel there was none of b and d from Table 1 in Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good?. Computational Brain & Behavior.
<https://doi.org/10.1007/s42113-023-00193-2>

Response 3: Addressed minimally in Items 5 and 8: discursive survival is reflected in usability by relevant scientific audiences, and minimising harm is included among theory-quality standards.

Comment 4: I think there's an argument for the theorist also having a distinct role in evaluating the work of other theorists. I don't think theorists work in silos (or think that they should not), and to be working with others, collaborating but also evaluating the work of peers should be an explicit role as it is in other domains of research.

Response 4: Addressed in Item 7 through comparative appraisal and in Item 8 through standards for judging theories, without adding a separate item.



Comment 5: Communicating the theory properly, also to non-theorists. Theories should eventually inform other psychologists' work, yet many have limited experience with theoretical work, especially formal modeling. It would be problematic if theoretical psychology develops independent from mainstream psychology. Just imagine a typical psychologists having to engage with a mathematical model relying on a set of differential equations. To me, there is an inherent challenge in presenting the necessary details of theoretical work and not overwhelming the readers one wants to reach. Navigating this challenge, in my opinion should be part of a theorist's tasks.

Response 5: Addressed in Item 5 by adding that theories should remain usable by relevant scientific audiences, while communication products themselves are treated as partly delegable.

Comment 6: This is a pretty complete list I think.

Response 6: No change required; this comment supports retaining the eight-item structure.

Comment 7: I would include the following: "Keep track of all activities, decisions, procedures, including all empirical evidence that has been conducted with the aim of testing a theory. This should include any failed attempts. To 'keep track of' also means to document everything that has been done, preferably in a FAIR way." (FAIR = findable, accessible, interoperable, reusable). Reason: necessary to enhance transparency and openness regarding the theorist's work.

Response 7: Addressed in Item 8 by adding documentation of assumptions, versions, and open questions, while FAIR infrastructure is treated as important but delegable.

Comment 8: See earlier points regarding the functions of theory Theories can be evaluated according to the same criteria for which they were originally developed eg description explanation prediction control fruitfulness

Response 8: Addressed across Items 7 and 8 by making theory appraisal include explanatory value, empirical interface, prediction, intervention relevance, usability, and potential harms.

Comment 9: '- I think it is already quite long!

Response 9: Addressed by retaining the eight-item structure and incorporating suggestions through minimal wording changes rather than adding new items



Important but delegable tasks

Survey question:

Are there any tasks or responsibilities that are important for the list of theorists' core responsibilities/tasks, but not essential for the theorist to perform personally and/or that can be delegated to other experts?

Comment 1: #1 the theorist must know what is being formalized, but does not necessarily have to code or mathematically formalize the model. The theorist task is to ensure that the model faithfully represents the theory. #2 the theorist should define the measurement problem - measurement experts could solve parts of the technical implementation (i.e., scale construction, item response modeling, reliability estimation, measurement invariance testing, and validation) #3 theorists design or guide theory-relevant research programmes, but they do not always have to run the empirical studies themselves

Response 1: Addressed in the introductory clarification and in Items 3, 5, and 8 by distinguishing non-delegable theoretical responsibility from delegable formal, measurement, and empirical execution.

Comment 2: Empirical evaluation can be outsourced to others.

Response 2: Addressed as an important but delegable task: empirical execution can be outsourced, while specifying evidential relevance remains part of theoretical responsibility.

Comment 3: Both formal proofs and programming spring to mind, but this might be too limited in how we see "theorist" if those to which they are delegated are not theorists.

Response 3: Addressed as important but delegable technical work: proofs and programming can be collaborative, while theoretical fidelity remains the theorist's responsibility.

Comment 4: I think attempting to find applications could be delegated, because I don't think the theorist is always the best person for the job of application. I personally see the theorist's work as being somewhat more fundamental in nature, and that they would be best working with others or delegating when it comes to translating their work into concrete application/implication territory.



Response 4: Addressed as an important but delegable task: application development can be collaborative or delegated, while deriving theoretical implications remains core.

Comment 5: Building infrastructure for archiving and reusing theory

Response 5: Addressed as important but delegable infrastructure work, while theory documentation itself is incorporated into Item 8.

Comment 6: I rated 3. Coordinating the relation between theory and measurement lower than the other tasks. I think it requires strong empirical experience to derive the most valid measurement techniques. People specializing in theoretical psychology may not necessarily have such experience. When point 2. before is done well, I suspect that measurement is already constrained to such an extent, that specific measurement proposal don't have to be made. Moreover, what if our empirical techniques improve over time. Then, this former proposal is outdated. Hence, I agree that point 3. is important, yet it could be delegated to experimentalists.

Response 6: Addressed in Item 3 by clarifying that technical measurement development can be collaborative or delegated, while defining the measurement problem remains core.

Comment 7: I find it very difficult to think about this in such an abstract way...

Response 7: Addressed by shortening the revised items and adding a division-of-labor clarification to make the abstract role boundaries easier to interpret.

Comment 8: No, I think most of this deserved to be in the theorist task description.

Response 8: No change required; this supports retaining the core items while still clarifying that technical execution may sometimes be collaborative or delegated.

Comment 9: * Many of the things in the list can (in my view) only be achieved in collaboration with other researchers -- for instance, judging what counts as a valid indicator for a construct (you may need a test developer, diagnostician, methodologist here); relating potential mechanisms to each other (you may need a psychologist for the intraindividual level, a sociologist for the macrosocial level, an anthropologist for the evolutionary level, etc.). So many of the things here cannot be done by the theorist alone. "Formulating theory-guided research programmes" (which was one point in task no. #8) could be delegated to other experts.



Response 9: Addressed in the introductory clarification and in Item 8 by framing research programmes as something the theorist helps design rather than necessarily executes alone.

Comment 10: Yes building tests that manifest good construct validity Meaning that the test empirically behaves in ways that align with key theoretical propositions postulated by the theory

Response 10: Addressed in Item 3: building tests can be delegated, but specifying how construct validity relates to theoretical propositions remains core.

Comment 11: Formalizing

Response 11: Addressed in Item 5 and in the delegability clarification: formalization may be useful, but its technical execution can be delegated.



General comments

Survey question:

Do you have any other comments or general assessments of the list of theorists' core responsibilities/tasks?

Comment 1: Some items are extremely dense. Good theory development often requires narrowing, rejecting or specifying expected failures. I would therefore consistently include criteria such as precision and parsimony

Response 1: Addressed by shortening the item revisions, adding precision and parsimony to Item 6, and adding expected failures to Item 7.

Comment 2: The theorist has the choice of retaining a theory after it has been "falsified." The principle of tenacity matters and even Popper agreed that theorists should not be hasty in giving up theories. Relatedly, Larry Laudan also proposed a logic of pursuit, which indicates when a research direction is worthy of allocation resources to.

Response 2: Addressed in Item 8 by adding that theorists may retain or revise theories rather than automatically discard them in response to anomalies.

Comment 3: Am I understanding correctly these are core tasks that are unique to theorists, correct? So any core tasks common to all are omitted?

Response 3: Addressed in the introductory clarification by framing the list as core theoretical responsibilities rather than all tasks shared by scientists generally.

Comment 4: Overall, I think the lists is very nice and helpful as a guide. However, beyond the task I think is missing (see above), my impression was that points 7. and 8., as currently formulated, are at least partly redundant. To me "deriving implications" and "makes explicit what evidence should count for or against which version of a theory" (both 7) are similar to "formulating a theory-guided research program" (8). Moreover, "evaluates how well theories account for existing evidence" (7) sounds very similar to "evaluates whether existing theories can account for known phenomena" (8). My suggestion would be to either merge these two points or demarcate them more.



Response 4: Addressed by demarcating Item 7 as derivation, scope mapping, and comparative appraisal, and Item 8 as integration, revision, documentation, and programme design.

Comment 5: I was wondering each time: should a theorist do *all* this? When can you consider yourself a theorist?

Response 5: Addressed in the introductory clarification by stating that the list describes core responsibilities, not tasks every theorist must personally execute.

Comment 6: I think it's pretty good!

Response 6: No change required; this comment supports retaining the eight-item structure.

Comment 7: I do like that list very much. But my feeling after reading the list of task was: For some of these tasks, a "division of labor" may be helpful or even necessary. See my comment above: some tasks require expert knowledge that the "theorist" may not have. In addition, I was wondering whether a "theorist" may sometimes take the role of a "critical friend", who, in adversarial collaboration with a subject-matter researcher, works towards refining a theory specification up to the point where the theory is actually testable. Doing this in an "adversarial collab" style may help prevent the tendency to immunize your theory-testing activity...

Response 7: Addressed in the introductory clarification and in the delegability responses by distinguishing division of labor from non-delegable theoretical responsibility.

Comment 8: Be sure to include that theories can be developed for and evaluated in diverse settings including naturalistic experimental and clinical And using a variety of study designs To evaluate theories of constructs typically construct valid measures are required Evaluation of cons

Response 8: Addressed in Items 3 and 8 by retaining measurement validity as a theoretical concern and broadening future work to observational, experimental, interventional, formal, and measurement work.

Comment 9: Looks great, just quite long already.

Response 9: Addressed by retaining the eight-item structure and making the revised wording more compact.